AI-ENABLED CYBER INCIDENT & SAFETY WEB PORTAL FOR DEFENCE

ABSTRACT

In the modern digital era, cybersecurity has become a critical concern for organizations, governments, and defence sectors. With the increasing dependence on digital communication systems, cloud technologies, and network-based infrastructures, the risk of cyber threats has grown significantly. Cyber attacks such as phishing, malware infections, ransomware, data breaches, and unauthorized system access pose serious threats to sensitive information and operational security.

Defence organizations are particularly vulnerable to cyber attacks because they manage highly confidential data, strategic communication networks, and national security infrastructure. A single cyber incident can disrupt operations, compromise classified information, and cause severe damage to national security systems. Therefore, it is essential to develop intelligent cybersecurity solutions that can assist in identifying, reporting, and responding to cyber threats efficiently.

Traditional cybersecurity monitoring systems often rely heavily on manual analysis and reactive approaches. This makes it difficult to handle the growing number of cyber incidents and detect threats quickly. As cyber attacks become more sophisticated, there is a need for advanced solutions that can provide automated threat detection, faster response mechanisms, and improved situational awareness.

The **AI-Enabled Cyber Incident & Safety Web Portal for Defence** is designed to address these challenges by providing a centralized web-based platform for reporting, analysing, and monitoring cyber incidents. The portal allows users to submit cyber incident reports through structured forms while enabling cybersecurity analysts and administrators to track and manage incidents through a dedicated command centre dashboard.

The system integrates Artificial Intelligence techniques to assist in analysing cyber threats and predicting their severity levels. The platform also provides real-time insights and analytics that help cybersecurity teams understand incident patterns and make informed decisions. The portal is developed using technologies such as **Python, Django framework, HTML templates, Tailwind CSS, and Chart.js**, ensuring a secure and scalable architecture while incorporating role-based authentication, email verification, and anonymous user tracking to maintain system security and data privacy.

PROBLEM STATEMENT

With the rapid growth of digital technologies and internet-based systems, cyber attacks have become increasingly frequent and sophisticated. Defence organizations and government institutions rely heavily on digital networks for communication, data storage, and operational coordination. However, this growing dependence on digital infrastructure has also made these systems highly vulnerable to cyber threats such as phishing attacks, malware infections, ransomware, data breaches, and unauthorized access attempts.

Many existing cybersecurity systems depend on manual monitoring and traditional security tools to detect and respond to cyber incidents. While these systems provide basic protection, they often struggle to handle the large volume of cyber threats generated in modern network environments. Manual analysis of cyber incidents can be slow, time-consuming, and prone to human error, which may delay the identification of critical threats and increase the risk of damage to sensitive systems.

Another major challenge is the absence of a centralized platform where cyber incidents can be reported, analysed, and monitored effectively. In many cases, suspicious cyber activities are either not reported promptly or are handled through fragmented systems, making it difficult for cybersecurity teams to track patterns and respond efficiently. This lack of coordination reduces the ability of organizations to detect emerging cyber threats and take preventive measures.

Additionally, traditional security systems often lack intelligent mechanisms to analyse incident data and determine the severity of cyber threats. Without the support of automated analysis and predictive techniques, it becomes difficult for cybersecurity professionals to prioritize incidents and respond quickly to high-risk situations.

Therefore, there is a need for an intelligent and centralized cybersecurity solution that can simplify cyber incident reporting, assist in threat analysis, and provide real-time monitoring of cyber threats. The proposed **AI-Enabled Cyber Incident & Safety Web Portal for Defence** aims to address these challenges by providing a secure web-based platform that allows users to report cyber incidents, enables administrators to monitor threats through a command centre dashboard, and utilizes intelligent techniques to support faster and more effective cyber incident management.

INTRODUCTION

Cybersecurity has become one of the most important concerns in the modern digital world. With the rapid growth of internet technologies, cloud computing, and digital communication systems, organizations increasingly rely on online infrastructure for their operations. However, this growing dependence on digital systems has also increased the risk of cyber attacks such as phishing, malware infections, ransomware, and unauthorized access attempts.

Defence organizations are especially vulnerable to cyber threats because they manage highly sensitive information, strategic communication networks, and national security infrastructure. Cyber attacks targeting defence systems can lead to serious consequences, including disruption of operations, data breaches, and exposure of confidential information. Therefore, it is essential to develop advanced cybersecurity systems that can help detect, analyse, and manage cyber incidents effectively.

Traditional cybersecurity solutions often rely on manual monitoring and reactive approaches to identify threats. As cyber attacks continue to evolve in complexity and scale, these traditional systems are often unable to respond quickly or analyse large volumes of incident data efficiently. This creates a need for intelligent systems that can assist cybersecurity professionals in identifying threats and making faster decisions.

The **AI-Enabled Cyber Incident & Safety Web Portal for Defence** is designed to provide a centralized platform where cyber incidents can be reported, monitored, and analysed efficiently. The portal allows users to report suspicious cyber activities, while cybersecurity analysts can track incidents through an administrative dashboard and analyse potential threats.

The system integrates modern web technologies such as **Python, Django framework, HTML templates, Tailwind CSS, and Chart.js** to create a secure and scalable platform. It also incorporates features such as role-based authentication, email verification, and data management mechanisms to ensure system security. By combining intelligent analysis with modern web technologies, the system aims to improve cyber incident response and strengthen cybersecurity preparedness.

OBJECTIVES

The main objective of the **AI-Enabled Cyber Incident & Safety Web Portal for Defence** is to develop a secure and intelligent platform that enables efficient reporting, monitoring, and analysis of cyber incidents. The system aims to improve cybersecurity management by providing a centralized environment where cyber threats can be identified, evaluated, and handled effectively.

One of the primary objectives of this project is to design a **centralized web portal for cyber incident reporting**. The system allows users to report suspicious cyber activities such as phishing attacks, malware infections, unauthorized access attempts, and other cyber threats through structured reporting forms. This helps in collecting cyber incident data in an organized and systematic manner.

Another important objective is to **assist cybersecurity analysts in monitoring and managing cyber incidents**. The system provides a dashboard that allows administrators to track incident reports, monitor security status, and analyse cyber threats efficiently. This helps security teams respond to incidents more quickly and effectively.

The project also aims to **integrate intelligent techniques for threat analysis and risk assessment**. By analysing incident data and identifying potential patterns, the system can help determine the severity of cyber threats and support faster decision-making in cybersecurity operations.

Additionally, the system focuses on ensuring **secure access and user authentication** through role-based login systems, email verification, and controlled user access. These features help maintain system security and protect sensitive information.

Overall, the objective of the proposed system is to strengthen cyber incident management, improve response efficiency, and enhance cybersecurity awareness within defence environments

PROPOSED MODEL

The proposed system is an **AI-Enabled Cyber Incident & Safety Web Portal for Defence** designed to provide a centralized platform for reporting, analysing, and managing cyber incidents. The system aims to improve cybersecurity preparedness by enabling users to report cyber threats and allowing security analysts to monitor and respond to incidents through an organized web-based interface.

The portal provides a **secure incident reporting mechanism** where users can submit details about suspicious cyber activities such as phishing attempts, malware infections, fraudulent messages, and unauthorized system access. The information submitted through the reporting system is stored securely in a database and can be accessed by authorized administrators and cybersecurity analysts for further investigation.

Fig 1: Working model of the AI-Enabled Cyber Incident & Safety Web Portal for Defence showing user reporting, AI threat analysis, dashboard monitoring, and administrative control.

The proposed system also incorporates **intelligent threat analysis capabilities** that assist analysts in evaluating reported incidents and determining their severity levels. By analysing incident details and identifying possible patterns, the system can help security teams prioritize threats and take appropriate actions quickly.

Another important component of the system is the **Cyber Command Centre Dashboard**, which allows administrators to monitor cyber incidents and system activities in real time. The dashboard provides visual analytics such as total incidents reported, resolved cases, pending investigations, and overall system risk levels. This helps security teams gain better insight into the cybersecurity status of the organization.

The system also includes **secure authentication and role-based access control** to ensure that only authorized users can access sensitive information. Features such as email verification, session management, and anonymous user tracking mechanisms help maintain system security and protect user data.

Overall, the proposed system aims to create an intelligent and user-friendly cybersecurity platform that improves cyber incident reporting, enhances threat monitoring, and supports faster response to cyber threats in defence environments.

KEY FEATURES

The **AI-Enabled Cyber Incident & Safety Web Portal for Defence** provides several important features that help improve cyber incident reporting, monitoring, and threat analysis. These features are designed to support cybersecurity analysts and administrators in managing cyber threats more efficiently.

One of the major features of the system is the **Cyber Incident Reporting Module**. This feature allows users to report suspicious cyber activities such as phishing attempts, malware attacks, fraudulent messages, or unauthorized access. The system provides structured forms where users can submit incident details and relevant information, which helps in collecting and organizing cyber threat data effectively.

Another important feature is the **AI-Based Threat Analysis** capability. The system assists cybersecurity teams in analysing reported incidents and identifying potential threat patterns. By evaluating the information submitted by users, the system can help determine the severity level of the incident and provide useful insights for further investigation.

The system also includes a **Cyber Command Centre Dashboard** that provides administrators with a centralized view of cyber incidents. The dashboard displays important statistics such as total incidents reported, resolved cases, pending investigations, and overall risk levels. Visual analytics tools help security teams monitor system activity and identify cybersecurity trends more easily.

In addition, the portal provides **Role-Based Access Control and Secure Authentication** to ensure system security. Only authorized users can access specific features of the system based on their assigned roles. Features such as email verification, secure login mechanisms, and session management help protect sensitive information and maintain system integrity.

Overall, these features work together to create a secure, intelligent, and user-friendly cybersecurity platform that supports effective cyber incident management in defence environments.

SYSTEM ARCHITECTURE

The system architecture of the AI-Enabled Cyber Incident & Safety Web Portal for Defence represents the overall structure of the system and the interaction between different modules. The architecture is designed to ensure secure data handling, efficient incident reporting, and effective cyber threat monitoring. The system follows a modular design where different components work together to provide a complete cybersecurity management platform.

The architecture begins with the User Interface Layer, which allows users to interact with the system through a web portal. Users can access the platform using a browser and submit cyber incident reports through structured forms. The interface is developed using HTML templates and Tailwind CSS, providing a user-friendly and responsive experience.

Figure 2: System Architecture of the AI-Enabled Cyber Incident & Safety Web Portal for Defence

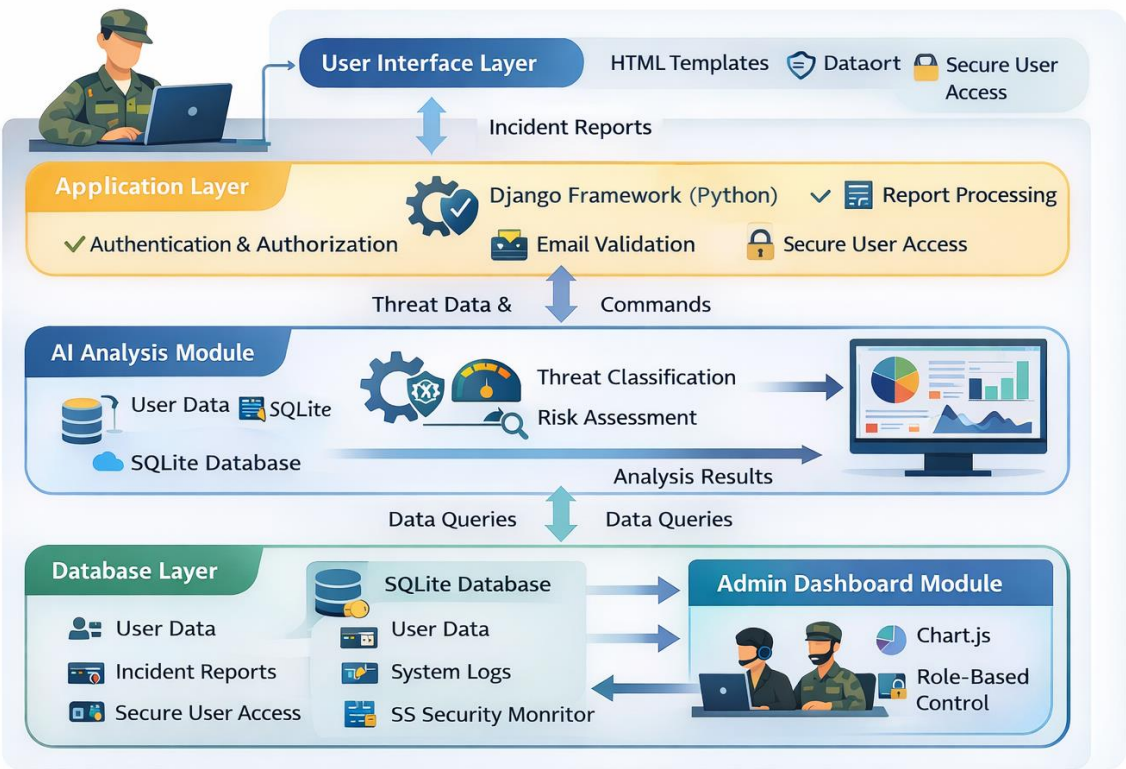


Figure 2: System Architecture of the AI-Enabled Cyber Incident & Safety Web Portal for Defence

The next component is the Application Layer, which is responsible for handling system logic and processing user requests. This layer is implemented using the Python programming language and the Django framework. It manages user authentication, incident report processing, data validation, and communication between different modules of the system.

The AI Analysis Module is responsible for assisting in the evaluation of cyber incidents. It analyses the information provided in incident reports and helps identify potential threats and risk levels. This module supports cybersecurity analysts in understanding possible attack patterns and prioritizing incidents based on their severity.

The system also includes a Database Layer, where all system data is stored securely. The database stores user information, incident reports, system logs, and analysis results. SQLite is used during development to manage and organize the stored data efficiently.

Finally, the Admin Dashboard Module allows administrators and cybersecurity analysts to monitor reported incidents through visual analytics and statistical reports. Tools such as Chart.js are used to display incident trends, risk levels, and system activity, helping analysts make informed decisions and respond to cyber threats effectively.

Overall, the system architecture integrates multiple modules including user interaction, application processing, threat analysis, and data management to create a secure and scalable cybersecurity platform.

TECHNOLOGIES USED

The AI-Enabled Cyber Incident & Safety Web Portal for Defence is developed using modern web technologies and tools that ensure security, scalability, and efficient system performance. The technologies used in the development of this project are described below.

Python

Python is used as the core programming language for backend development. It handles the main system logic, data processing, and communication between different components of the application.

Django Framework

Django is used as the web framework to manage server-side operations such as request handling, routing, authentication, and application logic. It also provides built-in security features and efficient database management.

HTML (Django Templates)

HTML templates are used to design the structure of the web pages and create the user interface for interacting with the system.

Tailwind CSS

Tailwind CSS is used to design a modern and responsive user interface. It helps in styling the web pages and improving the overall appearance of the portal.

JavaScript

Java Script is used to add interactivity and dynamic functionality to the web pages, allowing users to interact with the system smoothly.

Chart.js

Chart.js is used to create visual analytics on the admin dashboard. It displays incident statistics, charts, and other graphical representations of cyber incident data.

SQLite Database

SQLite is used as the database for storing system data such as user information, incident reports, and system logs.

Django Authentication System

The Django authentication system is used to manage user login, authentication, and access control within the system.

Custom User Model with Roles

A custom user model is implemented to support role-based access control, allowing administrators and users to access different features based on their roles.

Django ORM

Django ORM is used for database interaction and data management, allowing developers to work with the database using Python code.

Email OTP Verification

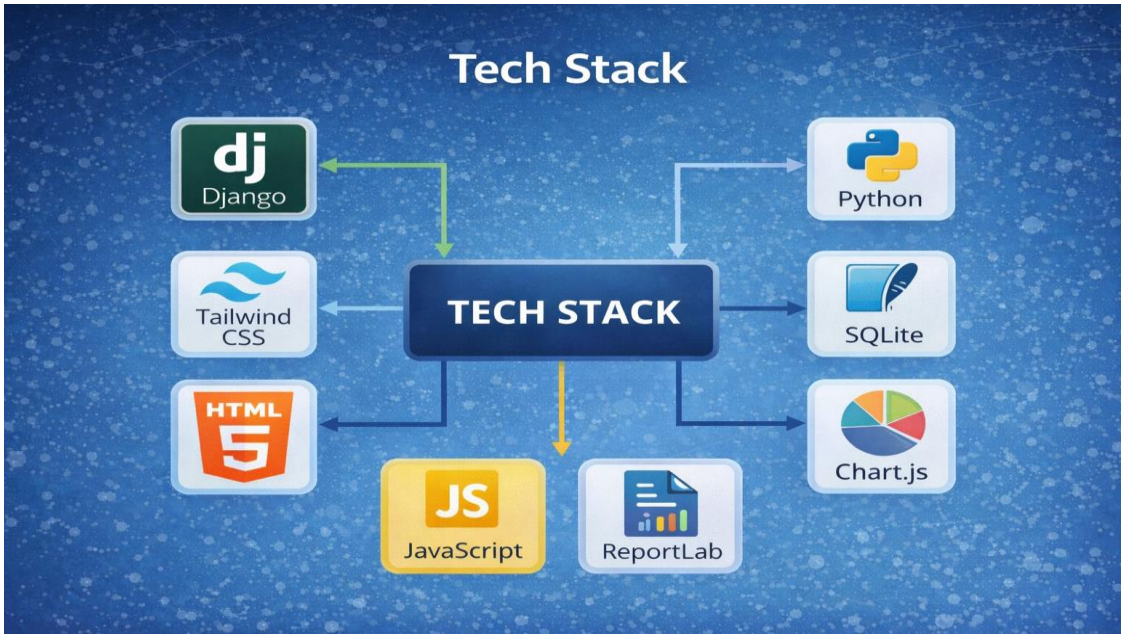
Email-based OTP verification is used to enhance security during user authentication and ensure that only verified users can access the system.

UUID-based Anonymous Tracking

UUID tracking allows anonymous users to report cyber incidents while maintaining privacy and secure identification within the system.

Report Lab

Report Lab is used for generating PDF reports within the system when require



IMPACT OF THE SYSTEM

The **AI-Enabled Cyber Incident & Safety Web Portal for Defence** provides an efficient and centralized platform for reporting and monitoring cyber threats. By allowing users to submit cyber incident reports through a structured system, the platform helps organizations identify and respond to potential cyber attacks more effectively.

The system also improves the ability of cybersecurity analysts to monitor incidents through a dedicated command centre dashboard. Visual analytics and incident statistics help analysts understand cybersecurity trends, evaluate potential risks, and take timely actions to prevent further threats.

Another important impact of the system is the reduction of manual workload for cybersecurity teams. By organizing incident data and providing intelligent insights, the platform helps analysts focus on critical threats and improves the overall efficiency of cyber incident management.

In addition, the platform increases cybersecurity awareness by providing users with a secure channel to report suspicious activities. This encourages proactive reporting and helps organizations detect cyber threats at an earlier stage.

CONCLUSION

Cybersecurity has become a critical challenge for organizations and defence systems due to the increasing number of cyber attacks and digital security threats. The **AI-Enabled Cyber Incident & Safety Web Portal for Defence** provides an intelligent and centralized solution that supports efficient cyber incident reporting, monitoring, and threat analysis.

The system integrates modern web technologies such as **Python, Django, HTML templates, Tailwind CSS, and Chart.js** to create a secure and scalable web platform. Through features such as incident reporting, analytics dashboards, and role-based access control, the system helps cybersecurity teams manage cyber threats more effectively.

Overall, the proposed system enhances cybersecurity preparedness by improving cyber incident management, supporting faster response to threats, and providing better visibility into cybersecurity risks.

SYSTEM DEPLOYMENT

The developed system has been successfully deployed and can be accessed through the following link:

DeploymentURL:

http://44.200.53.171:8000/

The deployed platform demonstrates the working functionality of the **AI-Enabled Cyber Incident & Safety Web Portal for Defence**, including cyber incident reporting, dashboard monitoring, and secure user authentication.